

From Assistance to Human Capability

A Readout-Genesis Architecture for Proactive AI, Unequal Life Conditions, and Opportunity Conversion

Human-owned problems, barrier typing, adaptive scaffolding, Human Return, and world-side recognition

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ORCID 0009-0005-3861-0626 • v1.1 — glosa-checked edition of v1.0, 6 September 2026 original

Status. K0 conceptual and measurement architecture with an empirical research agenda. The paper is rooted in the Readout Genesis contract and treats Human Capability Advancement (HCA) as a candidate domain translation, not as a new root and not as an established psychological law. It uses the Problem-First Dialogue Conversion Protocol, Human Learning as Epistemic Architecture, Choice Begins Before Choice, and CTSA Human Return as internal programme anchors. External literatures on scaffolding, adaptive instruction, self-determination, the capability approach, situated cognition, skill acquisition, human-AI reliance, job-market signaling, and child-centred AI are attached as convergence partners, rivals, measurement resources, or defeaters; none is imported as a root premise. No exact human-domain quotient or calibrated universal HCA scale is claimed. Licence CC BY 4.0.

Edition note. This is glosa-checked edition v1.1 of the founder's v1.0 (6 September 2026 original). Changes are limited to: (1) citation-lineage repair — the two working-specification self-citations (Readout Genesis Whitepaper; Information Epistemic Foundation) now carry a resolvable anchor (git repository/commit/blob, or the deposited record that carries the cited content) instead of an unanchored “working technical specification” label, and the six self-citations to since-superseded internal papers now carry their deposited Zenodo DOI and current version string (§4 Table 1; in-text labels “DCP v1.2”→“DCP v1.3” and “After Labour v1.1”→“v1.2”); (2) renaming §4.1 from a priority-claiming section heading to “Contribution boundary” and rewording the three sentences that carried priority-adjacent phrasing, including in negated form, per the programme's no-priority-language rule; (3) explicit [Definition]/[Open] tags added to the boxed non-collapse laws, candidate policies, and estimand equations, matching the tagging convention the v1.0 original already used for the Life-Capital Context Vector and the Human Capability Advancement Principle; (4) reference [37] replaced with a direct citation to the primary statute (Labour Protection Act B.E. 2541) after the cited Ministry of Labour guidance page could not be independently resolved. No section, equation, table, principle, invariant, study, rival, removal condition, or numbered claim was added, removed, renumbered in content, or otherwise reworded. The v1.0 PDF remains the anchor document of record; this edition is a citation-accurate, tier-labelled copy for deposit, not a superseding claim.

Central claim. When durable human development is an agreed goal, AI need not remain purely reactive. Starting from a human-owned live problem, it may proactively diagnose whether an observed difficulty is best read as a knowledge or skill gap, a language/tool/resource constraint, a network/credential/permission barrier, or another opportunity bottleneck; generate non-equivalent capability or opportunity routes; ask for human endorsement; provide adaptive support; progressively withdraw decisive assistance; test what returns unaided to the human; and connect retained capability to world feedback and real opportunity. The design target is therefore not maximum AI usage or maximum assisted output, but advancement that remains human-owned, survives AI removal where retention matters, and can be converted into action or opportunity under unequal life conditions.

ABSTRACT

Most AI assistants are evaluated at the point of assistance: answer quality, task accuracy, speed, productivity, or user satisfaction. These are important outcomes, but they do not establish that a human became more capable. Nor do they reveal whether a failure to act reflects a missing skill or a structural barrier such as language, time, tool access, network, credential, permission, or market recognition. This paper proposes a Readout-Genesis-native architecture for proactive human capability advancement. The proposal begins from a retained difference and a human-owned live problem, then inserts a barrier-typing stage before any training prescription. AI generates candidate capability and opportunity routes but does not silently own the user's goal. An endorsed route can then receive adaptive scaffolding, deliberate practice, assistance fading, unaided Human Return testing, novel transfer, world feedback, and—when relevant—a credential, portfolio, network, or institutional bridge.

The architecture separates objects that are often collapsed: observed difficulty from skill deficit; resources from access; access from control; assistance from retained capability; capability from opportunity; credential from capability; market legibility from human worth; and proactive support from agenda ownership. Unequal life conditions are represented as a context vector rather than assumed away, which changes both causal identification and policy interpretation. A worked Southern Thailand scenario shows why a low-resource child with strong real-world capability may remain institutionally illegible, why a formally schooled child can still become passively AI-dependent, and why the strongest pathway is likely to combine foundational education, human-owned problem inquiry, supervised AI scaffolding, real projects, Human Return, and credential/opportunity bridges.

External literature does not derive the model. Scaffolding and expertise-reversal evidence constrain assistance; self-

determination evidence constrains autonomy; the capability approach pressures the resource-to-opportunity transition; situated cognition and skill-acquisition work constrain transfer; human-AI studies motivate separation of assisted and retained performance; signaling theory constrains market recognition; and UNESCO/UNICEF guidance constrains child deployment. The residual contribution is a typed end-to-end architecture that asks not only what AI can do for a person now, but what the person can later do, what barriers still block conversion, and whether the route remains corrigible and human-owned.

Keywords: human capability; proactive AI; Readout Genesis; human-AI interaction; adaptive scaffolding; Human Return; unequal opportunity; capability approach; autonomy; learning; credential; child-centred AI.

1 THE PROBLEM: AN ASSISTANT CAN SOLVE THE TASK WITHOUT ADVANCING THE PERSON

Generative AI makes assistance cheap, fast, fluent, and increasingly available. That creates an immediate design temptation: optimize for the quality of the joint answer. Yet the empirical record already shows why that target is incomplete. Human-AI combinations do not exhibit universal synergy; a preregistered meta-analysis of 106 experiments found average human augmentation relative to humans alone but no average synergy relative to the better of human or AI, with decision tasks especially fragile [1]. In high-school mathematics, unrestricted generative-AI access improved practice performance but was followed by worse unaided exam performance than a no-AI condition, whereas guardrailed tutoring largely removed that learning cost [2]. Conversely, field evidence in customer support shows that AI can also facilitate durable worker learning, especially among less experienced workers [3]. The sign is therefore open: AI use may build capability, rent capability, or do both.

A second problem appears before learning. AI can influence what becomes salient, what is treated as a plausible explanation, and which topic is asked about next. Repeated biased AI interaction has been shown to shift later human perceptual, emotional, and social judgments, often without users fully recognizing the influence [4]. A protocol concerned with human development must therefore protect both the entry agenda and the returning human state.

A third problem is distributional. Two people given the same model do not possess the same effective opportunity. They may differ in language, foundational literacy, free time, device access, safety, health, mobility, mentor networks, credential routes, and institutional recognition. A design that reads every failure as a skill deficit risks training the person harder when the actual bottleneck is external. The central question of this paper is therefore:

Research question. Under what architecture may AI proactively advance human capability while preserving human-owned ends, distinguishing structural barriers from personal deficits, and separating assisted performance from durable Human Return?

2 READOUT GENESIS IS THE RAIL, NOT A METAPHOR

The proposal begins from the Readout Genesis constitutional discipline. The root does not contain a built-in object called skill, capability, motivation, education, credential, labour market, or human flourishing. Such terms are admissible only after a declared domain translation. The retained root state is written schematically as

$$S_n = (G_n, \Lambda_n, T_n), \quad (1)$$

with finite transition

$$S_{n+1} = F(S_n, u_n, c_n, T_n). \quad (2)$$

The candidate HCA domain must therefore satisfy a declared weld, at least approximately,

$$q_{\text{HCA},n+1}^\# \circ F_n = F_{\text{HCA},n} \circ q_{\text{HCA},n} \quad (3)$$

with corresponding readout and invariant conditions [5, 6]. The first non-collapse is constitutional [Definition]:

$$S_n \neq Z_{\text{HCA},n} \neq D_{\text{HCA},n}. \quad (4)$$

The retained root state, a candidate human-capability representation, and an empirically discovered sufficient quotient are different objects. Formal coherence in the HCA adapter would not establish that the psychological interpretation is true. The current status is therefore a candidate domain architecture with explicit defects and empirical gates, not a closed human science.

Readout Genesis also supplies two disciplines that become decisive here. First, access is bounded by reader, language, tool, permission, and context; access is not raw possession of reality [6]. Second, unresolved claims should specify the least-cost next record expected to discriminate the active uncertainty rather than silently converting ambiguity into a label. These rules will govern both barrier diagnosis and capability testing.

3 INTERNAL PROGRAMME LINEAGE: THE NATIVE RIVER BEFORE ALLIES ARE ATTACHED

The present paper does not derive its central objects from external educational theory. It inherits a sequence of already-developed internal interfaces and then asks whether they can be welded into a capability-advancement domain.

The native river is therefore

- Retained Difference → Human Readout
 → Live Problem
 → Barrier Readout
 → Candidate Routes
 → Human Endorsement
 → Adaptive Scaffold (5)
 → Practice
 → Withdrawal
 → Human Return
 → Novel Transfer
 → World Feedback
 → Opportunity Conversion.

The remainder of the paper formalizes this river and only then maps external literatures onto the locations where they have genuine explanatory or measurement authority.

4 EXTERNAL LITERATURE AS ALLIED RAILS, RIVALS, AND DEFEATERS

Scaffolding, adaptive tutoring, autonomy support, deliberate practice, knowledge transfer, capability conversion, and signalling are cited here as established literatures; this paper's contribution boundary is the typed ordering and non-collapse discipline across a larger human-AI capability-conversion process, not a claim about those literatures themselves. Those literatures are mature. The residual contribution is the typed ordering and non-collapse discipline across a larger human-AI capability-conversion process.

4.1 Contribution boundary

A reviewer should reject any claim that this paper invented adaptive tutoring, fading, transfer of responsibility, autonomy support, or conversion factors. The contribution here is narrower and specific: a Readout-Genesis-native architecture that (i) begins from a human-owned live problem rather than a fixed curriculum, (ii) types barriers before prescribing training, (iii) distinguishes capability routes from opportunity routes, (iv) makes proactive support subordinate to human endorsement, (v) uses assistance withdrawal and Human Return as explicit boundaries, (vi) returns the person to world-side feedback, and (vii) separately audits whether retained capability becomes institutionally legible and practically available.

5 CANDIDATE HCA DOMAIN STATE AND DEFECT DISCIPLINE

Let i index a human agent at finite step n . A candidate HCA state is

$$Z_{HCA,i,n} = \langle P_{i,n}^{\text{live}}, K_{i,n}^{\text{life}}, B_{i,n}^{\text{bar}}, C_{i,n}^{\text{cand}}, C_{i,n}^{\text{live}}, h_{i,n}, R_{H,i,n}^{\text{return}}, \Omega_{i,n}^{\text{real}} \rangle. \quad (6)$$

Here P^{live} is the endorsed live problem or explicit exploration goal; K^{life} is a life-context bookkeeping vector; B^{bar} is a typed barrier ledger; C^{cand} contains candidate capability/opportunity routes; C^{live} the subset endorsed as practical development routes; h the assistance/scaffold state; R_H^{return} the retained human return readout; and Ω^{real} a world-side opportunity/recognition readout.

This is a candidate representation, not a claim that the listed variables are minimal or sufficient. A serious implementation must report a defect vector

$$\varepsilon_{HCA} = \text{Def}(\tilde{q}_{HCA} \circ F, F_{HCA} \circ \tilde{q}_{HCA}, O_{HCA}, \text{Inv}_{HCA}), \quad (7)$$

and must be willing to delete variables when they fail to add reproducible explanatory or intervention value.

6 UNEQUAL LIFE CONDITIONS: CAPABILITY CANNOT BE IDENTIFIED FROM THE CONVERSATION ALONE

The user and AI do not meet on a blank plane. Define a Life-Capital Context Vector [Definition] as

$$K_{i,n}^{\text{life}} = \langle E^{\text{econ}}, F^{\text{base}}, L^{\text{lang}}, D^{\text{digital}}, T^{\text{disc}}, H^{\text{health}}, M^{\text{mob}}, N^{\text{mentor}}, C^{\text{cred}} \rangle_{i,n}. \quad (8)$$

The labels denote economic resources, foundational literacy/numeracy access, language bridge, device/connectivity, discretionary time, health/nutrition/sleep conditions, mobility/safety, mentor/network access, and credential pathway. This vector is not a universal cardinal measure of human capital and does not imply that these dimensions exhaust life conditions. It exists to prevent the causal fiction that people begin with equal conversion conditions.

The Readout-native point is stronger than “context matters.” The same world event can produce different accessible traces under different language, tools, permissions, and contexts [6]. The allied capability literature independently makes a similar distinction: resources are means, while substantive opportunities depend on conversion factors [13]. The present architecture preserves that convergence without importing the capability approach as a root ontology.

Three non-collapse laws follow [Definition]:

$$\text{Resources} \neq \text{Access}, \quad \text{Access} \neq \text{Capability}, \quad (9)$$

$$\text{Capability} \neq \text{Realized Opportunity}.$$

$$\text{Access}(z) \neq \text{Control}(z). \quad (10)$$

$$\text{Equal AI Access} \not\Rightarrow \text{Equal Capability Conversion}. \quad (11)$$

A child with a shared phone, weak Thai literacy, family work obligations, limited mobility, and no credential bridge is not equivalent to a child with private device access, uninterrupted schooling, free study time, and a mentor merely because both can open the same chatbot.

7 LIVE PROBLEM ENTRY AND BARRIER READOUT

RG-HCA inherits the Problem-First entry condition from DCP v1.3. A live problem is a presently consequential unresolved difference in the person's life, work, understanding,

Table 1: Internal programme lineage. Each prior work supplies a typed interface; the present paper restates the load-bearing object so it remains standalone.

Prior programme object	Retained result	Role in RG-HCA
Readout Genesis / Information Epistemic Foundation From Problem to Hypothesis	Retained difference precedes named semantics; domain claims require typed translation, provenance, selection policy, decisive records, and defect accounting. Practical inquiry can begin from a retained consequential mismatch; reachable alternatives differ from practically accessible alternatives.	Constitutional rail, no-import rule, barrier-readout discipline, and policy audit. [5, 6] Live Problem entry and non-equivalent candidate generation. [7]
How Humans Should Converse with AI v1.3	Human-development dialogue begins from human-owned topic entry, uses minimum sufficient friction, and closes through Human Return and world feedback.	Reactive dialogue protocol that RG-HCA extends with a proactive capability layer. [8]
Human Learning as Epistemic Architecture Choice Begins Before Choice	Problem-context, context-fit tools, workflow, deliberate practice, real problems, and feedback turn understanding into corrigible competence. Physical possibility, structural feasibility, live possibility, choice, enactment, and evaluator visibility must not collapse.	Competence engine after a capability route is endorsed. [9] Structural-barrier and opportunity layer; capability expansion is not the same as realized choice. [10]
CTSA Human Return	Concept, Tool, Skill, and Alternative Return require unaided reconstruction, selection, execution/transfer, or non-equivalent route generation.	Session and delayed return readout; gain/loss/provenance/warrant ledger. [11]
After Labour / Human Conversion (v1.2)	Human position depends on claim, exit, live alternatives, agency, retained capability, social reproduction, and recognition rather than output alone.	Macro boundary: capability that cannot be converted into standing or opportunity may remain economically invisible. [12]

Table 2: Global literature attached to the Genesis-native rail. No external school is treated as a root premise.

Rail location	Allied literature	What it establishes or pressures	Role here
Live problem / goal	Problem framing; goal articulation [14, 15]	Problems and goals are themselves consequential representations, not neutral inputs.	Convergent/Rival
Barrier vs opportunity	Sen/Capability tradition; conversion factors [13, 16]	Resources do not automatically become real opportunities; personal, social, and environmental conversion conditions matter.	Convergent; rival
Situated use Learner-state estimation Scaffold	Situated cognition [17] Knowledge tracing [18]	Knowledge and competence are partly constituted by activity, context, and culture of use. Interaction histories can be used to estimate skill mastery and personalize sequence or material.	Convergent Close rival; operational ancestor
Adaptivity	Wood–Brunner–Ross; scaffolding review [19, 20]	Assistance can be contingent, temporary, faded, and transferred back to the learner.	Convergent; no separate claim
Autonomy / structure	Expertise-reversal meta-analysis [21] Self-determination and autonomy-structure meta-analysis [22]	More support can help novices yet burden experts; assistance must depend on prior knowledge and domain. Structure and autonomy support can reinforce rather than negate one another.	Convergent; Defeater
Skill consolidation	Anderson; deliberate practice; retrieval practice [23–25]	Exposure and declaration are insufficient; proceduralization, effortful retrieval, feedback, and repeated execution matter.	Convergent
Human-AI learning	Bastani; Brynjolfsson [2, 3]	AI can either impair later unaided performance or facilitate durable learning depending on design and context.	Convergent; Defeater
Reliance and friction	Buçinca; Vasconcelos; de Jong; Tian [26–29]	Critical engagement can reduce overreliance but imposes cognitive and adoption costs.	Convergent; burden constraint
Human-centric control policy	Offline RL for human-centric objectives [30]	Assistance policies can optimize objectives beyond immediate accuracy; learning is harder to optimize than accuracy.	Close rival
Human-AI system performance Feedback loops	Vaccaro–Almaatouq–Malone [1] Glickman–Sharot [4]	Human+AI is not automatically synergistic and effects vary by task. Repeated AI interaction can alter later human judgments, sometimes outside awareness.	Convergent; Defeater
Market recognition	Spence signalling [31]	Employers infer latent productivity from visible signals; observed credentials can matter independently of actual capability.	Convergent; justifies return audits
Child deployment	UNESCO; UNICEF [32–34]	Age-appropriate human oversight, safety, privacy, participation, and rights are necessary constraints.	Convergent; market-legibility layer Governance constraint

evidence relation, or environment [7, 8]. Problem-first is a default for development, not a prohibition on curiosity, play, art, or basic inquiry.

The proactive extension begins only after the problem is human-owned. The next question is not “Which skill should AI teach?” but “What currently blocks this problem from

becoming solvable or the desired route from becoming live?”
Define a typed barrier ledger

$$B_{l,n}^{\text{bar}} \subseteq \{\text{Knowledge, Skill, Language, Tool, ResourceTime, Network, Credential, Permission, Opportunity, Unknown}\}. \quad (12)$$

The load-bearing non-collapse is [Definition]:

$$\text{Observed Difficulty} \neq \text{Skill Deficit}. \quad (13)$$

A wrong diagnosis has real costs. If the active barrier is language, repeated skill drills may punish the learner for a translation problem. If the barrier is device access, teaching advanced digital workflow does not create hardware. If the barrier is a missing credential, more private competence may remain invisible to an employer. If the barrier is permission or safety, encouragement to “try harder” can be irresponsible.

Barrier typing must therefore remain probabilistic and falsifiable. Let $U_{i,n}^{\text{diag}}$ be admissible low-risk diagnostic actions: asking a clarifying question, requesting a small unaided attempt, checking language comprehension, confirming tool access, or observing a real-world failure point. A candidate decisive-record policy is [Open]:

$$u_{i,n}^{\text{diag}*} = \arg \max_{u \in U^{\text{diag}}} [I_{G_B}(u) - \lambda_C \text{Cost}(u) - \rho \text{Risk}(u)], \quad (14)$$

where $I_{G_B}(u)$ is the expected information/discrimination gain among active barrier hypotheses. Equation 14 is Open: it adapts the Genesis decisive-record discipline to a human-development domain and should be replaced if a simpler rule performs as well.

8 CANDIDATE ADVANCEMENT ROUTES: CAPABILITY AND OPPORTUNITY ARE DIFFERENT OBJECTS

Once the barrier ledger is explicit, AI may generate candidate advancement routes

$$C_{i,n}^{\text{cand}} = \text{Gen}(P_{i,n}^{\text{live}}, B_{i,n}^{\text{bar}}, K_{i,n}^{\text{life}}, R_{H,i,n}^{\text{return}}). \quad (15)$$

The candidate set may include learning routes (numeracy, writing, diagnosis, spreadsheet use), access routes (translation, tool, mentor), and opportunity routes (portfolio, certificate, apprenticeship, network introduction). Raw idea count is not the target. Two routes are distinct only if they differ in required capability, barrier relaxed, predicted world consequence, verification route, or institutional pathway.

The AI is allowed to be proactive about proposing routes, but not about silently making them the person’s goals. Define

$$C_{i,n}^{\text{live}} = \{c \in C_{i,n}^{\text{cand}} : \text{Endorse}_i(c) = 1\}. \quad (16)$$

For an adult, endorsement means informed, revisable agreement. For a child, especially a young child, endorsement is insufficient by itself; deployment also requires age-appropriate assent plus guardian/teacher or institutional oversight under a child-centred governance policy [32, 34].

This yields three further non-collapse laws [Definition]:

$$\text{Proactive Suggestion} \neq \text{Human Goal Ownership}, \quad (17)$$

$$\text{Capability Advancement} \neq \text{AI Goal Authority}. \quad (18)$$

$$\text{Scaffolding} \neq \text{Control}. \quad (19)$$

The external autonomy-support literature is useful precisely here. Structure and autonomy support can reinforce one another rather than form a zero-sum pair [22]. That literature constrains how AI may help; it does not decide which life goal the person should endorse.

9 THE PROACTIVE HUMAN CAPABILITY ADVANCEMENT POLICY

A reactive assistant waits for the user to request help. RG-HCA permits a narrower proactive action: after a human-owned problem and barrier readout, AI may surface a development opportunity that the user has not explicitly requested, provided it makes the reason visible and preserves refusal.

Let $U_{i,n}^{\text{adv}}$ contain admissible interventions such as a hint, worked comparison, mini-practice task, retrieval prompt, tool demonstration, mentor/credential suggestion, or real-world challenge. A candidate domain-level policy is [Open]:

$$u_{i,n}^{\text{adv}*} = \arg \min_{u \in U^{\text{adv}}} \left[\mathbb{E} \|\text{Res}_{i,u}^{\text{cap}}\|^2 + \lambda_B B^{\text{use}}(u) + \rho R_u + \mu D_u + \xi A_u + \psi O_u - \nu V_u \right], \quad (20)$$

subject to

$$\begin{aligned} \text{AgendaOwnership} &= 1, & \text{Transparency} &= 1, \\ \text{Endorsement} &= 1, & \text{Safety/Governance} &= 1. \end{aligned} \quad (21)$$

Res^{cap} is a declared residual between the capability needed for the endorsed route and current demonstrated return; B^{use} is usability burden; R_u is task or safety risk; D_u dependency risk; A_u autonomy loss or agenda displacement; O_u opportunity cost; and V_u expected cross-problem transfer value. The weights are local policy parameters, not universal human utilities. Any empirical use must predeclare and sensitivity-test them.

Equation 20 is deliberately close in form to the existing Genesis epistemic-action policy that trades residual reduction against cost, risk, repair loss, and value [6]. The added human-development terms are a domain proposal, not a claim of root derivation.

Human Capability Advancement Principle [Open]. When durable human development is an endorsed goal, AI may proactively identify and propose a reachable capability or opportunity expansion, but it should remain subordinate to human-owned ends, prefer the least burdensome intervention likely to produce durable return, and withdraw decisive support as the person demonstrates independent capability.

10 ADAPTIVE SCAFFOLDING: USE WHAT THE LEARNING SCIENCES ALREADY KNOW, BUT KEEP THE BOUNDARIES

The local mechanism of support is the same one used in the cited scaffolding literature. Scaffolding has long emphasized contingent assistance, fading, and transfer of responsibility [19, 20]. Recent meta-analytic evidence on expertise reversal reinforces the need for adaptivity: low-prior-knowledge

learners benefit from more support on average, whereas high-prior-knowledge learners can be harmed by excess assistance [21]. The model therefore rejects a fixed amount of friction.

Let $h_{i,n}$ denote the level of decisive assistance. Rather than prescribe a universal update law, RG-HCA uses a monotonic candidate rule [Open]:

$$\text{Stable Unaided Return} \uparrow \Rightarrow h^{\text{decisive}} \downarrow \quad (22)$$

unless task difficulty, stakes, or domain risk rises. Failure after reduced support can justify temporary re-scaffolding. This allows assistance to be high for a novice and low for an expert without treating either as a moral preference.

For skill-building tasks the process is

$$\begin{aligned} \text{Attempt} &\rightarrow \text{Minimal Sufficient Scaffold} \rightarrow \text{Feedback} \\ &\rightarrow \text{Reattempt} \rightarrow \text{Fading} \\ &\rightarrow \text{Unaided Execution} \rightarrow \text{Novel Transfer.} \end{aligned} \quad (23)$$

This is consistent with established work on declarative-to-procedural skill acquisition, deliberate practice, and retrieval practice [23–25]. The internal Human Learning architecture places these mechanisms after problem-context and tool selection rather than treating them as a complete account of human development [9].

11 HUMAN RETURN: THE AI SHOULD KNOW WHEN TO BECOME LESS NECESSARY

A system can produce a superior joint answer while leaving little retained capability with the person. RG-HCA therefore inherits CTSA Human Return:

$$R_H^{\text{return}} = \langle C, T, S, A \rangle, \quad (24)$$

where C is Conceptual Return, T Tool-selection Return, S Skill-execution Return, and A Alternative-generation Return. The decisive evidence is unaided reconstruction, selection, execution/transfer, or generation under a declared criterion [11].

The primary non-collapse remains [Definition]:

$$\Delta \text{Performance}_{\text{AI}} > 0 \not\Rightarrow \Delta R_H^{\text{return}} > 0. \quad (25)$$

A retained false belief is still retained; retention is not warrant. A new fast procedure may be a wrong procedure; skill is not direction. A complete audit therefore records gain, loss, provenance, warrant, and retained direction rather than only gross gain [11].

The design slogan “AI should make itself unnecessary” must also be bounded. It is appropriate when the target is capability that the human has reason to retain. It is not a universal prohibition on delegation. For trivial formatting, repetitive administration, or other low-value procedures, permanent delegation can be rational. The correct statement is [Open]:

$$R_H^{\text{return}} \uparrow \Rightarrow h^{\text{decisive}} \downarrow \quad (26)$$

for capability-sensitive tasks, unless task difficulty, stakes, or domain risk rises.

12 WORLD RETURN: CAPABILITY THAT NEVER LEAVES THE CONVERSATION REMAINS UNPROVEN

Human Return establishes what returned with the person, but an action-relevant live problem is not closed by verbal reconstruction. DCP v1.3 already requires return to the world [8]. Let $a_{i,n}$ be a human-selected action and $\delta_{i,n+1}^{\text{world}}$ the returned record:

$$R_{H,i,n}^{\text{return}} \rightarrow a_{i,n} \rightarrow \delta_{i,n+1}^{\text{world}} \rightarrow Z_{\text{HCA},i,n+1}. \quad (27)$$

The returned record may confirm the route, expose a new barrier, show that the problem was misframed, or reveal that the acquired skill does not transfer under real constraints. Situated cognition provides an external ally for this step: knowledge and competence cannot always be abstracted from the activities and contexts in which they are used [17].

13 OPPORTUNITY CONVERSION: THE MARKET MAY NOT BE ABLE TO READ THE CAPABILITY

A further boundary appears after Human Return. The person may now be capable yet still lack a viable social or economic route. Define a world-side opportunity readout

$$\begin{aligned} \Omega_{i,n}^{\text{real}} = G_O(R_{H,i,n}^{\text{return}}, K_{i,n}^{\text{life}}, \\ \text{Cred}_{i,n}, \text{Net}_{i,n}, \text{Perm}_{i,n}, \\ \text{MarketReadout}_n). \end{aligned} \quad (28)$$

This is not a welfare utility function and not a validated scalar. It is a reminder that realized opportunity is a joint readout of retained capability and conversion conditions. Two distinctions are essential [Definition]:

$$\text{Credential} \neq \text{Capability}, \quad (29)$$

$$\text{Market Legibility} \neq \text{Human Worth}. \quad (30)$$

Spence’s signalling model is an obvious external neighbour because employers must infer latent productivity from visible signals under uncertainty [31]. RG-HCA uses this literature only to motivate a legibility problem: capability that is not represented by a recognizable credential, portfolio, demonstration, reference, or network may fail to become labour-market opportunity. The remedy can therefore be an opportunity bridge rather than more skill training.

A full advancement route may end with

$$\text{Skill} \rightarrow \text{Evidence of Skill} \rightarrow \text{Recognition} \rightarrow \text{Opportunity}, \quad (31)$$

but the same chain can fail at any transition.

14 WORKED SCENARIO: THREE CHILDREN, UNEQUAL STARTING CONDITIONS, AND THE AGE-15 BOUNDARY

The following scenario is an analytical stress test, not a forecast of individual children. It is motivated by a real equity

problem in Thailand's deep south. UNICEF/NSO MICS reporting found very low basic reading rates among children aged 7–8 in Narathiwat, Pattani, and Yala relative to the national average, and higher upper-secondary non-attendance in parts of the region [35]. A recent reintegration initiative for NEET youth in four southern provinces reported that many participants returned to education, employment, or training after targeted outreach and skill/opportunity support [36]. These facts make the region an appropriate example of why life conditions and opportunity bridges belong inside the model.

Define three stylized children who are age seven in 2026 and age fifteen in 2034 (Table 3). The causal warning is [Definition]:

$$\text{Outcome}_C - \text{Outcome}_A \neq \text{Effect}_{HCA}, \quad (32)$$

because C receives more schooling, credential access, and other life conditions. The cleaner comparison is B versus C under matched baseline conditions. A serves a different scientific question: whether an HCA system can reduce the amount of potential lost when starting conversion conditions are poor.

Age fifteen is also not equivalent to full labour-market access. Thai law prohibits employment below age fifteen and imposes additional restrictions for workers under eighteen, including limits on night work, overtime, and hazardous work [37]. The age-15 target should therefore be interpreted as readiness for safe entry-level work, supervised apprenticeship, project income, or vocational transition—not as a recommendation to end education. The strongest pathway suggested by this scenario is therefore not “AI instead of school” but

$$\begin{aligned} &\text{Schooling} + \text{Human-Owned Inquiry} \\ &\quad + \text{Supervised Adaptive AI} \\ &\quad + \text{Projects} + \text{Human Return} \\ &\quad + \text{Opportunity Bridges.} \end{aligned} \quad (33)$$

15 CHILD AND ADULT GOVERNANCE: PROACTIVITY WITHOUT PATERNALISM

Proactivity increases both benefit and manipulation risk. The architecture therefore distinguishes adult deployment from child deployment. For adults, the minimum governance bundle is:

$$\begin{aligned} &\text{Purpose} + \text{Transparency} + \text{Refusal} \\ &\quad + \text{Revision} + \text{Data Minimization.} \end{aligned} \quad (34)$$

The system must disclose why it is proposing a capability route, link the route to the human-owned problem, allow refusal without penalty, and permit the user to revise or delete the development goal.

For children, especially younger children, a stronger gate is required:

$$\begin{aligned} &\text{Child Assent} + \text{Adult Oversight} \\ &\quad + \text{Age-Appropriate Design} \\ &\quad + \text{Privacy} + \text{Safety} \\ &\quad + \text{No Opaque Persuasion.} \end{aligned} \quad (35)$$

UNESCO's GenAI guidance calls for human-centred, age-appropriate use, privacy protection, and age limits for independent conversations with GenAI platforms [32]. UNICEF's current strategy and guidance emphasize human autonomy and oversight, inclusion, transparency, privacy, safety, development, and children's participation [33, 34]. A seven-year-old RG-HCA deployment should therefore be mediated by a trusted adult or institution; child assent alone is not treated as sufficient governance.

The system must also know when not to be proactive. No advancement intervention is required for every conversation. The default should remain minimum sufficient friction, with proactivity triggered only when the user has opted into development, a repeated barrier is visible, or stakes justify a capability check.

16 MEASUREMENT ARCHITECTURE: WHAT WOULD COUNT AS ADVANCEMENT?

A full HCA evaluation must separate at least six outcome families.

1. *Immediate assisted performance.* Did the person solve the current task faster or better with AI?

2. *Human Return.* After decisive assistance is removed, can the person reconstruct the concept, choose the tool, execute the skill, or generate a non-equivalent alternative? [11]

3. *Transfer.* Does the retained capability work on a sufficiently novel problem, not merely the practiced item?

4. *Autonomy/agenda integrity.* Can the person identify the original goal, explain why the proposed route was accepted, reject a route, and revise the goal without the system steering them back?

5. *Burden and adoption.* Time, cognitive load, interruption cost, satisfaction, abandonment, and repeated-use willingness matter because an epistemically ideal protocol that nobody uses has low field value.

6. *Opportunity conversion.* Does the capability become usable in a real institution, project, credential pathway, apprenticeship, or labour-market context? Is capability legible to relevant gatekeepers?

A candidate net advancement record is therefore a measurement/estimand definition [Definition]:

$$\begin{aligned} A_i^{HCA} = \langle &\text{Gain}_{CTSA}, \text{Loss}, \text{Transfer}, \\ &\text{Ownership}, \text{Burden}, \text{BarrierChange}, \\ &\text{OpportunityChange}, \text{Provenance}, \\ &\text{Warrant} \rangle. \end{aligned} \quad (36)$$

No scalar score is required. For policy comparison, a Pareto report is preferable unless a value weighting is explicitly declared.

17 TIER-ONE EMPIRICAL PROGRAMME

The theory should earn its complexity through experiments that permit it to lose components.

Table 3: Illustrative A/B/C scenario. The table is a theory stress test, not a forecast and not an estimate of treatment effects.

	A: low life capital + RG-HCA	B: formal school + passive AI	C: formal school + RG-HCA
Starting conditions	Interrupted/weak schooling, lower language/credential access, limited devices/time/network; supervised problem-first AI and mentor support added.	More continuous schooling and credential path; AI mainly used as answer generator.	Same formal schooling/credential path as B plus supervised problem-first HCA, projects, fading, Return tests.
AI relation	AI proposes routes from real problems; may build high practical capability but cannot manufacture institutional recognition by itself.	User asks, AI answers; immediate task performance can rise without deliberate return testing.	AI assists, challenges, teaches when endorsed, fades, and pushes transfer to projects/world evidence.
Age-15 capability	Potentially strong local problem solving/portfolio relative to formal attainment, but foundational gaps remain possible.	Stronger curriculum/credential legibility than A, but practical problem framing and unaided AI-complementary capability are not guaranteed.	Highest plausible joint position because foundational schooling, credential, portfolio, and Human Return can accumulate together.
Market legibility	Weakest if no M.3-equivalent/recognized signal; community/project markets may see capability better than formal HR systems.	Formal systems can read the credential even if capability is ordinary.	Credential and portfolio make capability more legible; still constrained by age and sector.
Research meaning	Tests whether scarce life resources can be converted into retained capability, not whether AI “beats school.”	Baseline for formal schooling with passive AI.	Cleanest contrast with B for estimating the added value of HCA under matched schooling/life conditions.

17.1 Study 1: reactive DCP versus proactive HCA

Recruit adults across low- and high-prior-knowledge strata and assign them to: (A) direct-answer AI; (B) problem-first reactive DCP; (C) proactive HCA with barrier typing and capability suggestions; (D) no-AI. Use real or realistic problem domains in which assistance can be removed. Primary outcomes: immediate performance, delayed CTSA Return, novel transfer, autonomy/ownership, burden, and reliance calibration. The critical question is whether proactive HCA adds delayed capability beyond a strong reactive protocol.

17.2 Study 2: skill-only tutor versus barrier-aware navigator

Create tasks in which failure is experimentally caused by different bottlenecks: missing knowledge, missing skill, language comprehension, tool access, or credential/permission constraints. Compare a tutor that assumes skill deficit with a barrier-aware condition. The main outcome is correct barrier identification and intervention selection. If barrier typing adds no predictive or intervention value beyond ordinary mastery models, the barrier layer should be simplified.

17.3 Study 3: adaptive scaffold and fading

Cross prior knowledge with high versus low assistance and test whether HCA’s assistance policy reproduces the expertise-reversal pattern rather than imposing a one-size policy. Evaluate whether fading based on demonstrated Return outperforms fixed fading schedules.

17.4 Study 4: longitudinal supervised youth pilot

A child trial must be institutionally supervised and age-appropriate. A plausible design in a multilingual, resource-variable context would compare matched learners receiving (B) formal schooling plus ordinary/passive AI support with (C) formal schooling plus supervised problem-first HCA. Measure foundational literacy/numeracy, school engagement, CTSA Return, real-project completion, language bridge, portfolio evidence, and credential continuity over 6–24 months. Do not use employment at age fifteen as the primary endpoint; use safe vocational readiness, transfer, and continued learning.

17.5 Study 5: market-legibility experiment

Use employer or supervisor vignettes to orthogonally vary demonstrated capability, credential, portfolio evidence, and referral/network signal. Test whether identical capability receives different opportunity under different signals. This study operationalizes the distinction between capability and market legibility and tests whether a portfolio/credential bridge changes hiring or apprenticeship judgments.

17.6 Identification discipline

The child A/B/C scenario shows why baseline context cannot be ignored. Let Y be a declared outcome and HCA an intervention indicator. The target estimand is not a raw rank difference but a conditional contrast such as a measurement/estimand definition [Definition]:

$$\text{ATE}_{\text{HCA}}(x) = \mathbb{E}[Y(1) - Y(0) \mid K^{\text{life}} = x]. \quad (37)$$

Here x denotes a declared baseline life-context stratum; matching or covariate adjustment is a design strategy, not part of the estimand itself. Randomization is preferable where ethical and feasible; otherwise, the paper should report the limits of identification rather than convert association into “AI impact.”

18 RIVAL MODELS AND LEGITIMATE DEFEATERS

RG-HCA is not protected from simpler explanations.

Rival 1: ordinary adaptive tutoring is enough. Knowledge-tracing and intelligent-tutoring traditions already estimate learner mastery from interaction histories and adapt instruction [18]. If a well-designed tutor with mastery estimation, contingent support, and fading predicts all relevant delayed outcomes, the HCA architecture should collapse toward that simpler model.

Rival 2: proactive human-centric policy optimization is enough. Bućinca et al.’s human-centric policy framework is a close computational rival [30]. RG-HCA earns separate value only if human-owned problem entry, barrier typing, opportunity conversion, or Readout-native return accounting add reproducible information beyond optimized assistance.

Rival 3: the capability approach already supplies the whole theory. The capability tradition gives a powerful normative and conversion vocabulary [13, 16]. RG-HCA should not duplicate it. Its distinct burden is to specify a finite human-AI interaction pipeline with readout, barrier discrimination, scaffolding, removal, return, provenance, and empirical control policies.

Rival 4: Human Return is just ordinary transfer. If CTSA categories add no reliable information beyond standard knowledge, skill, transfer, and metacognitive measures, CTSA should be merged or removed [11].

Rival 5: proactive support creates paternalism or disengagement. If proactive suggestions reduce autonomy, increase abandonment, suppress exploration, or disproportionately burden low-literacy users, the triggering policy should become more restrictive or revert to reactive assistance.

Rival 6: opportunity bridges do not change realized opportunity. If credential, portfolio, network, or institutional routing add no stable predictive value after measured capability, the opportunity-conversion layer should be simplified.

19 FALSIFICATION AND REMOVAL CONDITIONS

The following findings would count against the architecture rather than being reinterpreted as support:

- barrier categories cannot be coded reliably across independent raters or add no intervention value;
- proactive HCA produces no delayed Return or transfer advantage over reactive DCP at comparable burden;
- the endorsement mechanism does not preserve agenda ownership or is routinely manipulated by framing;
- fading based on Return is no better than fixed assistance schedules;
- CTSA adds no incremental validity beyond ordinary delayed-performance measures;
- life-context variables do not moderate conversion outcomes in the claimed domains;
- opportunity/credential bridges do not improve any declared real-world conversion outcome;
- users reject the protocol under realistic time pressure so strongly that protective gains disappear;
- child deployment cannot satisfy privacy, safety, oversight, and age-appropriate participation constraints;
- a simpler rival explains the same outcomes with fewer assumptions.

An architecture that cannot lose variables becomes vocabulary rather than theory.

20 WHAT THE PAPER DOES NOT CLAIM

The paper does not claim that every person should constantly self-improve, that every conversation should become a lesson, or that AI should decide which life is worth living. It does not claim that low-resource children can substitute chatbot access for foundational schooling, stable adult relationships, nutrition, safety, or credentials. It does not claim

that credentials are fair measures of capability, only that institutions often read them. It does not claim that proactivity is always better than reactivity, that all AI use causes deskilling, or that all scaffolding should end in complete unaided performance. It does not claim an exact HCA quotient, a universal life-capital score, or a global labour-market forecast.

The strongest admissible claim at present is architectural: if the goal is durable human development, then a complete human-AI advancement system should distinguish human-owned ends, barrier type, candidate route, assistance, retained return, and opportunity conversion rather than collapsing them into current task success.

21 DISCUSSION: FROM AI AS ANSWER ENGINE TO AI AS SUBORDINATE CAPABILITY NAVIGATOR

The practical change proposed by RG-HCA is small in interface terms but large in objective. A conventional assistant asks, “What answer should I give?” A tutor asks, “What should this learner practice next?” A capability navigator asks a different sequence:

1. What human-owned problem or exploration goal is active?
2. What barrier is actually blocking progress?
3. Which non-equivalent capability or opportunity routes could relax it?
4. Which route does the person endorse?
5. What is the minimum sufficient assistance needed now?
6. When should decisive support fade?
7. What returned with the person after AI removal?
8. Does the retained capability transfer to a new problem and survive world feedback?
9. Is there an external bridge needed so the capability becomes a real opportunity?

This sequence prevents a subtle but consequential error: treating a user who repeatedly needs help as proof that the user needs more help. Sometimes the correct move is a short practice intervention. Sometimes it is a translation. Sometimes a mentor, tool, credential, safe place, or institutional route is the real missing object. The theory therefore turns “proactive AI” away from behavioural engagement and toward a falsifiable conversion problem.

Three design invariants summarize the proposal.

Human-end invariant. AI may be proactive about expanding human capability, but it must remain subordinate to human-owned ends.

Return invariant. When a capability is worth retaining, success is demonstrated by what the human can still do when decisive AI assistance is absent.

Structural-barrier invariant. When the barrier is structural, the intervention must change access or opportunity—not merely train the person harder.

22 CONCLUSION

Human-AI systems are moving from tools that answer toward systems that remember, recommend, plan, and increasingly act. A purely reactive model of assistance is therefore no longer the only relevant design target. Yet proactivity creates its own danger: a system capable of noticing

weaknesses and proposing development can also redefine the person's problem, substitute its own goals, pathologize structural barriers as personal deficits, or make dependence feel like progress.

The Readout-Genesis response is to type the transitions. A retained difference becomes a human readout; a human-owned live problem is separated from an AI-supplied agenda; observed difficulty is separated from barrier type; candidate routes are separated from endorsed routes; assistance is separated from Human Return; retained capability is separated from realized opportunity; and credential or market recognition is separated from human worth. External literatures then attach where they genuinely constrain the rail: scaffolding at assistance, self-determination at endorsement, capability theory at conversion, learning science at practice, human-AI evidence at reliance and return, signalling at recognition, and child-rights governance at deployment.

The paper's central empirical wager is therefore not that AI should "install skills." It is that a carefully governed AI can sometimes act as a subordinate capability navigator: it can identify a likely bottleneck, propose a reachable development or opportunity route, scaffold the person's own work, withdraw, test transfer, and help connect retained capability to the world. Whether this architecture actually produces more durable, more equitable, and more human-owned capability than strong reactive or tutoring baselines is an open question. The next step is to test it hard enough that parts of the architecture are allowed to fail.

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